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Week 6: Image Processing

Reading: Week 6 Lecture (30 min)

Ungraded Pledge: 6.1 Overview of Image Processing (5 min)

Ungraded Pledge: 6.2 Fourier Transforms (17 min)

Practice Assignment: 6.2 Fourier Transform Self-check Quiz (Submitted)

Reading: 6.3 Video: Convolution (10 min)

Ungraded Pledge: 6.3 Convolution Theorem (8 min)

Practice Assignment: 6.3 Convolution Theorem Self-check Quiz (Submitted)

Ungraded Pledge: 6.4 Image Filtering in Frequency Domain (14 min)

Practice Assignment: 6.4 Image Filtering in Frequency Domain Self-check Quiz (Submitted)

Ungraded Pledge: 6.5 Deconvolution (17 min)

Practice Assignment: 6.5 Deconvolution Self-check Quiz (5 min)

Reading: 6.6 Video: Convolution (10 min)

Ungraded Pledge: 6.6 Sampling Theory and Aliasing (13 min)

Practice Assignment: 6.6 Sampling Theory and Aliasing Self-check Quiz (10 min)

Discussion Prompt: Week 6 Questions and Feedback

Week 6 Quiz

Post Course Survey

Your grade: 100%

Next item →

6.4 Image Filtering in Frequency Domain

Self-check Quiz

1. If given a sharp image, how do we want to remove as blurred images will only have low frequency components?

1 / 1 point

Note: Band Pass Filter allows only a range of frequency to pass through and filters out any higher and lower frequency components

☐ Apply High Pass Filter

☐ Apply Band Pass Filter

☒ Apply Low Pass Filter

☐ None of the above

Ready to review what you've learned before starting the assignment? I'm here to help.

Help me practice

Let's chat

Assignment details

☒ Correct

Sharp image has high frequency components. We want to remove as blurred images will only have low frequency components.

Submitted

Oct 31, 1:40 AM PST

Download

Try again

2. How can we extract only edges from an image which also has only granular features in it?

1 / 1 point

Note: Band Pass Filter allows only a range of frequency to pass through and filters out any higher and lower frequency components

☐ Apply High Pass Filter

☒ Apply Band Pass Filter

☐ Apply Low Pass Filter

☐ None of the above

Like

Dislike

Report an issue

Correct

Edges have high frequency associated as compared to other image features, also noise have much higher frequency hence need a band pass filter to extract edges perfectly.

Submitted

Oct 31, 1:40 AM PST

View submission

See feedback

https://www.coursera.org/learn/cameraandimaging/assignment-submission/V3Qv6-4-image-filtering-in-frequency-domain-self-check-quiz/view-feedback

1/1